勒让德幂级数递推：偶次与奇次系数各自闭合，整数阶在有限项终止

偶链：$a_0 \to a_2 \to a_4 \to \cdots$

奇链：$a_1 \to a_3 \to a_5 \to \cdots$

$$a_{n+2} = \frac{n(n+1) - L(L+1)}{(n+1)(n+2)} a_n$$

$$P_0 = 1, \qquad P_1 = x$$

$$P_2 = \frac{1}{2}(3x^2 - 1)$$

$$P_3 = \frac{1}{2}(5x^3 - 3x)$$